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Effect of Curved Boundary Layer Fences on Aerodynamic Efficiency

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Increment in aerodynamic efficiency was observed at high angles of attack on a NACA 0012 wing with the use of curved boundary layer fences (CBLFs). Historically, boundary layer fences (BLFs) were installed on the upper surface of airplane wings to prevent flow separation. These traditional BLFs provide marginal benefits in post stall lift while preventing stall propagation along the spanwise direction of the wing. In this research, the conventional BLF was curved to influence the spanwise flow direction. Therefore, apart from preventing propagation of stall, the added benefits of the CBLF is that it changes the surface spanwise velocity which has implications in the wingtip vortex roll up process while also acting as a potential control surface. Before quantifying the effect of CBLFs on the wingtip vortex and as a control surface, the influence of CBLFs on wing performance is investigated experimentally in the University of Dayton Low Speed Wind Tunnel (UD-LSWT). Force based experiments were done to determine the changes in the wing performance with different configurations of CBLFs (upper or lower surface coverage, fractional spanwise coverage, spanwise position, and orientation of curvature). CBLF test configurations were compared to a baseline NACA 0012 wing as well as traditional, non-curved BLF configurations. Preliminary results indicate an increase in aerodynamic efficiency at angles of attack greater than 10º for all cases. The increment in aerodynamic efficiency is largely due to decreased parasitic drag coefficient and not increment in lift coefficient.

Nomenclature

AoA, α	=	Angle of attack [°]
AR	=	Aspect ratio
A	=	Axial force $[N]$
BLF	=	Boundary layer fence
C_D	=	Coefficient of drag
C_L	=	Coefficient of lift
CBLF	=	Curved boundary layer fence
D	=	Drag $[N]$
L	=	Lift $[N]$
N	=	Normal force $[N]$
ρ	=	Density $\left[\frac{kg}{m^3}\right]$
S	=	Surface area $[m^2]$
V_∞	=	Freestream velocity $[{}^{m}/_{s}]$

I. Introduction

1.1 Literature Review

During lift production, the air from the high-pressure region on the bottom surface of the wing curls towards the low-pressure region on the upper surface at the wingtip causing a spiraling vortex. These vortices induce downwash which in turn creates induced drag. Since every lift-generating wing causes these vortices, induced drag is inevitable. Passive wingtip devices such as end plates, winglets, and wingtip sails are currently being employed in full scale airplanes to reduce the strength of wingtip vortices. However they only result in drag reduction of few percent. The effects of induced drag are most prevalent during takeoff and landing where the lift production, hence the strength of wingtip vortices, is higher. Apart from higher induced drag, non-uniform flow separation across the span occurs in low aspect ratio (AR) wings which results in non-uniform lift distribution and pitching moments. Therefore,

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unconventional passive methods of improving aerodynamic efficiency under these specific conditions must be explored.

One method aimed at improving aerodynamic efficiency of aircraft is through the use of passive control devices known as boundary layer fences (BLFs), also known as stall fences or wing fences. Apart from being used to prevent stall from propagating across the span of a wing, numerous literatures have shown that BLFs provide aerodynamic benefits in the form of an increased lift coefficient [1], [2], [3], [4], [5]. Some early swept wing aircraft such as the Mikoyan-Gurevich MiG 15 and MiG 17 as well as the non-swept de Havilland Canada DHC-6 utilize BLFs in their design.

Typically, BLFs are added to an aircraft after design to improve lift and stall characteristics without significantly affecting other flight properties. For example, a 1953 research memorandum by Bray [6] of the National Advisory Committee for Aeronautics (NACA) discusses the effects of wing fences on the longitudinal stability of the F-86A swept-wing aircraft. This particular aircraft was selected because it experienced upwards pitch at mild lift coefficients which affected its high-speed maneuverability, particularly in the transonic Mach range of 0.87 to 0.93. To help mitigate these negative effects, several different types of BLFs were placed at multiple locations on the aircraft's wing and observed. Some BLFs covered the entire chord while some covered part of it. Some BLFs covered the leading edge of the wing and others did not. The results from this study showed less pitch-up in the Mach range of 0.87 to 0.93 which was attributed to the arrangement of BLFs.

There are a few theories as to why BLFs work to improve lift and stall characteristics of aircraft. Originally, BLFs were placed on the wing of the Messerschmitt Bf-109 fighter by German aerospace engineer Wolfgang Liebe [7] because the aircraft was experiencing stall at lower angles of attack (AoAs) than were desired. He determined that the reason for this was because of spanwise flow influencing flow separation on the upper surface of the wing. By adding a fence, the spanwise flow is blocked which turned out to be one of the common theories for why BLFs improve both stall and lift characteristics of aircraft [7]. However, spanwise flow along the wing will play a role in defining the performance of the CBLF as well. According to Garmann and Visbal [8], spanwise flow at the wingtips has been shown to affect the rollup of the wingtip vortex. This can be observed in **Figure 1** which is borrowed from the paper by Garmann and Visbal.

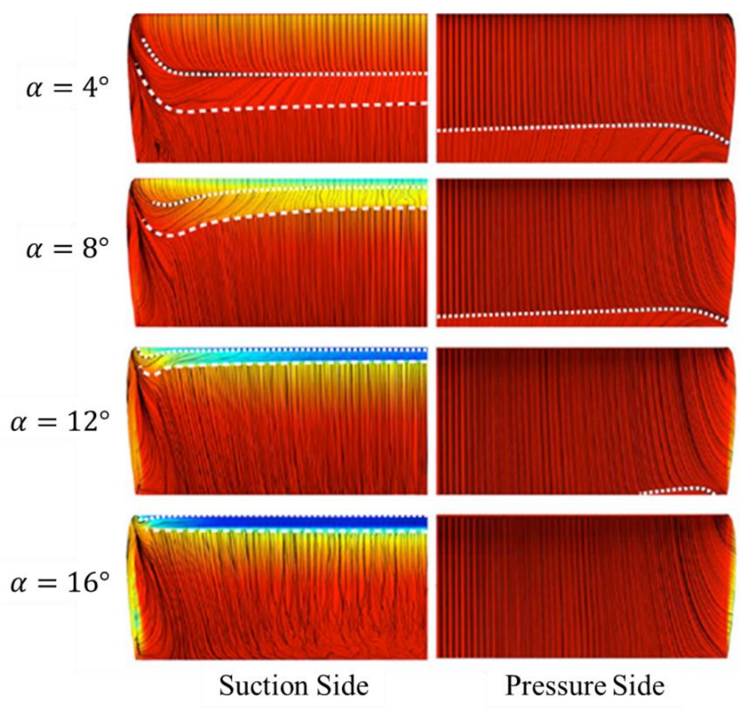

Figure 1: Surface pressure contours show spanwise flow contributing to wingtip vortex rollup on a NACA 0012 wing on the upper surface (left) and lower surface (right). ***Image borrowed from Garmann and Visbal [8].***

When spanwise flow is restricted, it may influence the pressure gradient at the wingtip, hence the strength of the wingtip vortex, which would affect induced drag. With this theory in mind, it would make sense that BLFs would be made just slightly higher than the boundary layer in order to minimize drag but still function the same. This isn't the case however since much experimentation has shown that BLFs need to be several times higher than the boundary layer or else they don't provide the same benefits to lift coefficient [7].

Another BLF theory deals with vortex generation. Through CFD analysis of a T-38 wing with a BLF located near the wingtip, Solfelt and Maple [9] found that two oppositely rotating vortices formed outboard of the fence and caused flow to stay attached near the wingtip. The T-38 has a swept wing and because stall initially occurs at the wingtips of swept wings, the BLF delayed stall from propagating across the span to the rest of the wing. Solfelt and Maple documented a 7% increase in coefficient of lift from the BLF. Also, the max lift coefficient occurred at an AoA of 13˚ as opposed to 12˚.

A recent study by Walker and Bons [10] provides an in-depth examination of the effects of passive BLFs as well as active flow control (AFC) wall-normal steady blowing slots that act as BLFs that can be actuated at different times during flight. These devices were tested on an AR 4.3 NACA 64_3-618 swept wing via force based experimentation, particle image velocimetry (PIV), and mini-tuft flow visualization. Results from the study showed a 14.3% increase in $C_{L\ \max}$, a 10º increase in stall angle, and increased lift performance between the AoA range of 18º and 28º by placing a traditional passive BLF on the upper surface of the swept wing. Research done by Salmi [3] and Pratt and Shields [4] showed that mounting several BLFs usually outperform single BLFs. It is hypothesized that the use of a single CBLF could replace several BLFs. In most BLF cases, they are placed only on the upper surface of the wing however recent research by Goodman and Gunasekaran [11] showed that the near-wake from the lower surface showed the same trend as the near-wake from the upper surface. This could be due to similar properties in the spanwise flow on both of these surfaces. Because of this, CBLFs were designed to cover the upper and lower surfaces of the NACA 0012 wing independently to determine variation between the different configurations.

As stated before, BLFs influence the spanwise velocity over the wing surface. Because of the curvature and increased wetted area of CBLFs, spanwise velocity is hypothesized to be influenced differently when compared to straight BLFs. If BLFs were made from some compliant structure, they could be actuated to adjust their curvature during flight and behave as a control surface. These morphing BLFs could have an effect on flight dynamics, particularly for small scale unmanned flight vehicles.

1.2 Objective

The objective of this research is to determine the changes in aerodynamic efficiency with the use of CBLFs of different fractional spanwise coverages, spanwise locations, and directions of curvature on a semispan AR 2 NACA 0012 wing. CBLFs were compared with straight BLFs as well as a baseline wing without any BLFs. Coefficients of lift and drag were used to quantify these effects. It is hypothesized that by placing a CBLF on the surface of a wing, spanwise flow is affected which in turn affects the lift distribution on the wing as well as formation, growth, and evolution of the wingtip vortex. Due to the increased cross-sectional area with the addition of CBLFs, it is hypothesized that the drag coefficient will increase when compared to the baseline but also be accompanied by increment in lift. The aerodynamic efficiency is expected to increase at higher AoAs.

II. Experimental Setup

2.1 Test Models

All 3-D rendering of the test wings was done using Dassault Systèms SolidWorks 2016-2017 Student Edition. A semispan AR 2 wing was designed and printed using a stereolithography apparatus (SLA) printer. The outline of the NACA 0012 airfoil with a chord length of approximately 101.94 mm was simply extruded 202.5 mm to achieve a semispan AR of 2. Two holes were extrude cut into the root of the wing in order to place threaded inserts that would allow for the wings to interface with the force sensor. Square slots were added onto the leading edge of the wings in order to secure the CBLFs and BLFs during testing as they featured inserts on their inner portions. Leading edge slots on the wings were spaced accordingly to accommodate all test configurations of the CBLFs and straight BLFs based on spanwise position and direction of curvature. Additional leading edge inserts were designed to cover the remaining slots that were not covered already so that there were no complications with the data caused by the open slots on the leading edge. **Figure 2** shows an isometric and front view of one of the wings that was designed for testing along with its dimensions. This wing was used to test the half span CBLF.

A NACA 0012 airfoil was used as the basis for the creation of two different CBLFs and one traditional straight BLF. All of the BLFs, both curved and straight, completely cover one surface of the wing and the leading edge while also covering 10% of the opposite surface. CBLFs and the BLF were approximately 6-7 mm in height from the wing surface depending on the spanwise location. The front of the CBLFs and BLF extends a quarter of the chord from the leading edge of the wing and the back extends a quarter of the chord plus 10 mm from the trailing edge. The reason for this additional 10 mm was to match with a previous CBLF design that covered both surfaces of the wing simultaneously. In this design, an additional 10 mm was needed to place a slot and insert on a flat portion of the CBLF near the trailing edge so that the upper and lower halves of the CBLF could snap into place over the wing. These dimensions are illustrated in a side view of a CBLF shown in **Figure 3**.

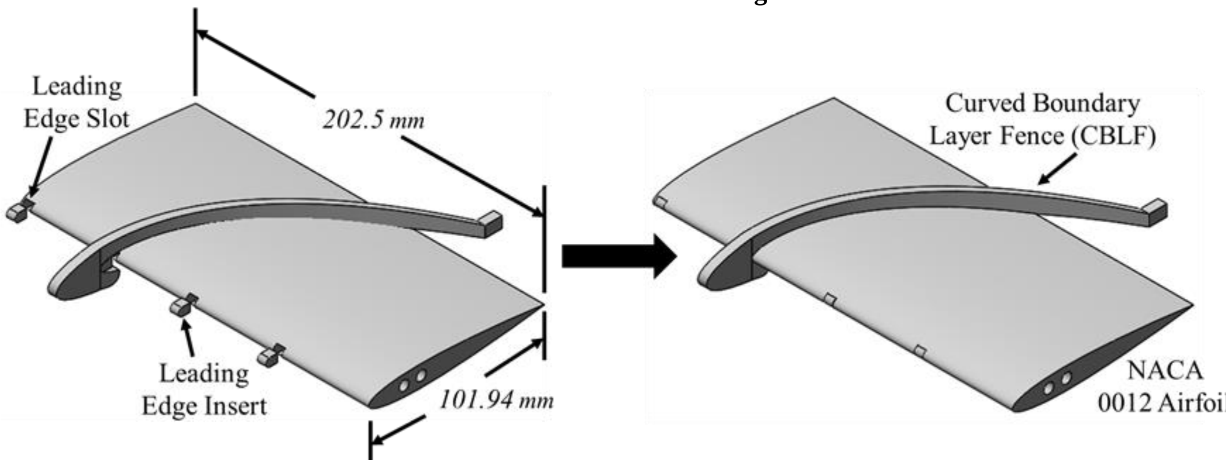

Figure 2: Semispan AR 2 test wing that was printed and used for force based testing of the half span CBLF. Leading edge slots were added to the design in order to mate with interior inserts on CBLFs and straight BLF to secure fences during wind tunnel testing.

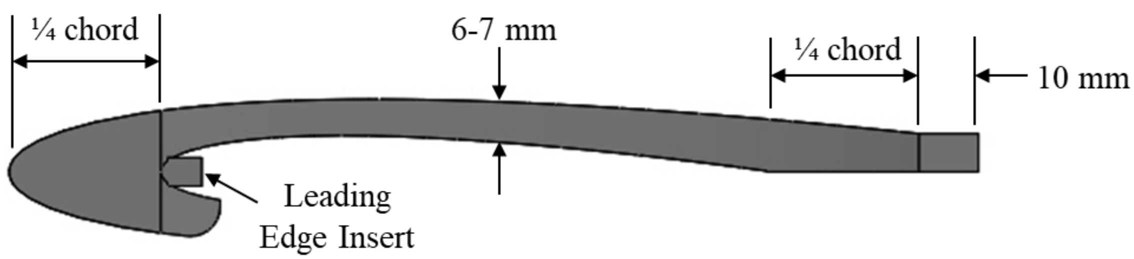

Figure 3: Side view of a CBLF. Leading-edge insert was designed to mate with leading-edge slots on the wings to secure CBLFs and BLF during wind tunnel testing.

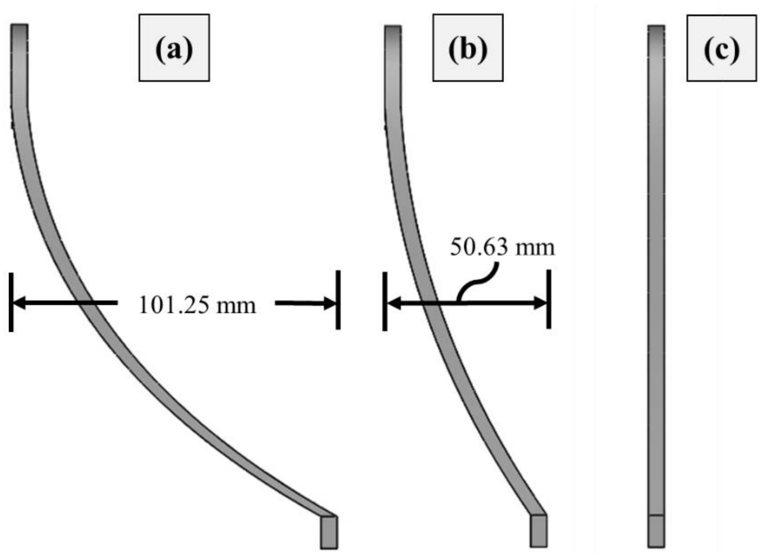

Figure 4: Top view of both CBLFs and the straight BLF to show the differences in spanwise coverage; (a) half span, (b) quarter span, (c) straight BLF (a.k.a. zero span).

The two CBLFs varied only in the fraction of the 202.5 mm semispan that they covered; half span (**Figure 4a**) and quarter span (**Figure 4b**). Both were 5 mm thick and their curvature was created using the spline function. The straight BLF, (**Figure 4c**) was also 5 mm thick and was designed in the same way as the CBLFs but with zero curvature. As stated earlier, force testing was performed on the wing with CBLFs to understand their effect on the aerodynamic performance. These results were compared to the baseline NACA 0012 wing without BLFs as well as the wing with the traditional straight BLFs for comparison. The details for this experimental setup are discussed in the section below.

2.2 Wind Tunnel

All experiments were conducted in the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The wind tunnel has an inlet contraction ratio of 16:1 and can be converted to an open or closed jet configuration with a freestream range of 3 m/s to 37 m/s. The open jet configuration was used in all aerodynamic force testing. The 76.2 cm x 76.2 cm inlet opens to a pressure sealed plenum and the 137 cm x 137 cm collector sends the rushed air through to the diffuser. A picture of the UD-LSWT and the setup for the force based testing is shown in **Figure 5**.

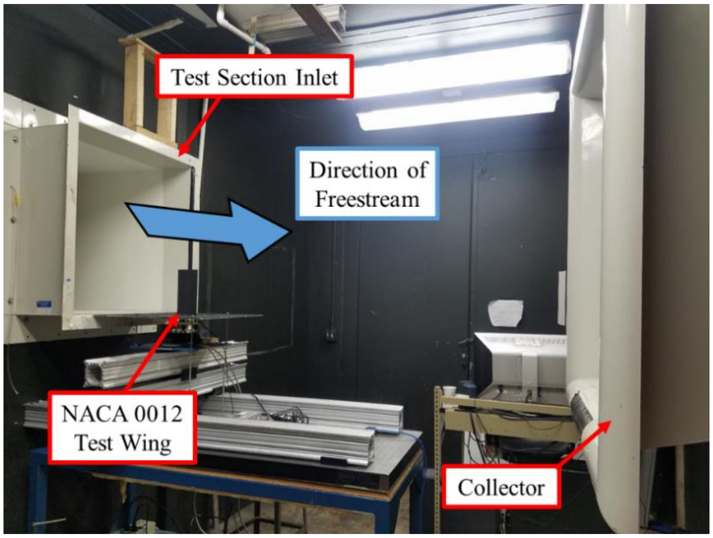

Figure 5: Setup of the baseline force based test in the UD-LSTW with the open jet configuration.

2.3 Aerodynamic Force Testing

All force based testing on the semispan AR 2 NACA 0012 wing was done at a Reynolds Number of approximately 250,000 (freestream velocity of 35 m/s) and at AoAs ranging from -5º to 15º.

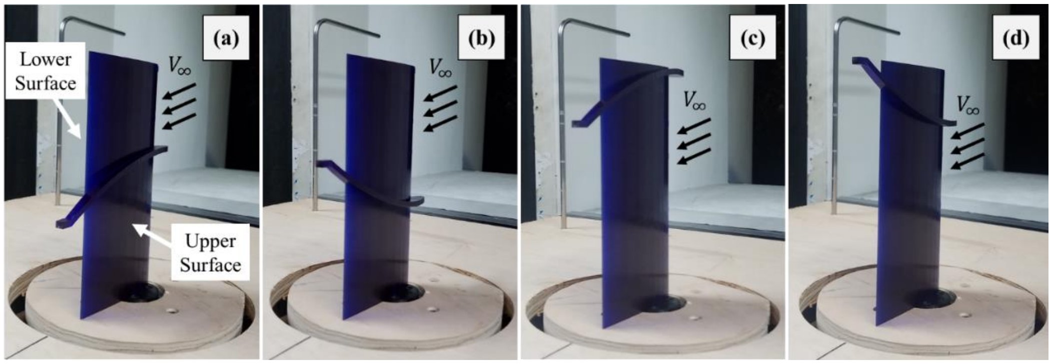

Figure 6: Upper surface quarter span CBLF test configurations; (a) middle inward, (b) middle outward, (c) wingtip inward, (d) wingtip outward. The same orientations and positions were done on the lower surface and for the half span CBLF yielding 16 total CBLF test configurations.

The two CBLFs were tested in configurations that differed in upper or lower surface coverage, spanwise location on the wing (middle or wingtip), and direction of curvature (inward or outward). To be clear, the CBLF configurations on either surface were as follows; middle inward (**Figure 6a**), middle outward (**Figure 6b**), wingtip inward (**Figure 6c**), and wingtip outward (**Figure 6d**). Again, both CBLFs were tested in these four configurations on both the upper and lower surfaces independently yielding 16 total CBLF test configurations. The straight BLFs were tested at different spanwise locations only but also on the upper and lower surfaces which yielded four total straight BLF test configurations. Including the baseline wing without CBLFs, there were 21 total cases. The test matrix for the force-based experiments is shown in **Table 1**.

Table 1: Test matrix for the force based experiments.

<table>
<tr><th>Airfoil</th><th>Semi-Span AR</th><th>Single Surface CBLF</th><th>Surface of Coverage</th><th>Spanwise Location</th><th>Direction of Curvature</th><th>Reynolds Number</th><th>AoA Range (degrees)</th><th>Trials</th></tr>
<tr><td rowspan="21">NACA 0012</td><td rowspan="21">2</td><td>Baseline</td><td>N/A</td><td>N/A</td><td>N/A</td><td rowspan="21">248,762</td><td rowspan="21">-5 to 15</td><td rowspan="21">1</td></tr>
<tr><td rowspan="8">Half Span</td><td rowspan="4">Upper</td><td rowspan="2">Middle</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="2">Wingtip</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="4">Lower</td><td rowspan="2">Middle</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="2">Wingtip</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="8">Quarter Span</td><td rowspan="4">Upper</td><td rowspan="2">Middle</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="2">Wingtip</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="4">Lower</td><td rowspan="2">Middle</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="2">Wingtip</td><td>Inward</td></tr>
<tr><td>Outward</td></tr>
<tr><td rowspan="4">Zero Span (Straight BLFs)</td><td rowspan="2">Upper</td><td>Middle</td><td rowspan="4">N/A</td></tr>
<tr><td>Wingtip</td></tr>
<tr><td rowspan="2">Lower</td><td>Middle</td></tr>
<tr><td>Wingtip</td></tr>
</table>

The ATI Mini-40 force transducer was used to measure the normal and axial forces on the wing which were later converted into lift and drag coefficients. The range and resolution for each direction of the sensor is shown in **Table 2**. The sensor was located underneath the splitter plate along with the Griffin rotary stage which was used to change the angle of attack. The rotary stage was controlled using Galil motion software. A schematic of force-based test setup is shown in **Figure 7**.

Table 2: Range and resolution for the Mini-40 force transducer.

Force/Torque Direction	F_x (N)	F_y (N)	F_z (N)	T_x (Nm)	T_y (Nm)	T_z (Nm)
Range	40	40	120	2	2	2
Resolution	1/100	1/100	1/50	1/4000	1/4000	1/4000

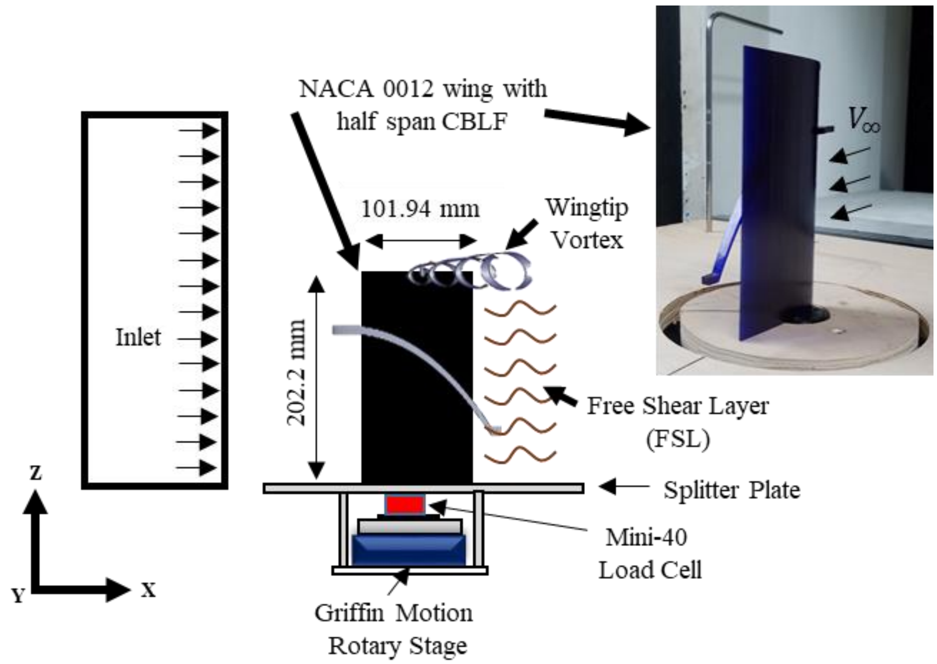

Figure 7: Schematic for the setup of the force based experiments shows all equipment used.

III. Results

3.1 Overview

Because 21 unique test cases were investigated in the force based experiments, the test results will be split up into three sections.

- Comparison of coefficients without any BLFs (baseline) and with straight BLF.
- Sensitivity of CBLFs to fractional spanwise coverage (quarter and half span coverage)
- Sensitivity of CBLFs to upper or lower surface coverage.

The last two parameters, spanwise location and direction of curvature, will not be discussed in this paper because the data shows that aerodynamic efficiency is not significantly affected by these two parameters.

The normal and axial forces recorded by the Mini-40 force transducer were used to calculate lift and drag forces using **Equation 1** and **Equation 2**. In these equations, N and A are the normal and axial forces respectively while α is angle of attack.

$$L = N\cos\alpha - A\sin\alpha \tag{1}$$
$$D = N\sin\alpha + A\cos\alpha \tag{2}$$

Once lift and drag forces were calculated, the coefficients of lift and drag could be calculated using **Equation 3** and **Equation 4** respectively. The same planform surface area S (0.021m^2) was used for the baseline and the wings with BLFs, both curved and straight, when calculating the lift and drag coefficients as it was assumed that the thickness of each CBLF and BLF was negligible. Further, density ρ (1.255 kg/m^3) and velocity V_∞ (35 m/s) values were the same for all cases.

$$C_L = \frac{2L}{\rho\, {V_\infty}^2 S} \tag{3}$$
$$C_D = \frac{2D}{\rho {V_\infty}^2 S} \tag{4}$$

3.2 Baseline and Straight BLF Results

The baseline NACA 0012 wing was tested at a Reynolds number of approximately 250,000 (freestream velocity of 35 m/s) and through an AoA range of -5º to 15º. The wing that was used for the straight BLF test configurations was used as the baseline wing. Two additional inserts were placed into the two leading edge slots so that the leading edge of the wing was smooth. Straight BLFs were tested to compare their performance to the CBLF test configurations but also to determine the influence of BLFs on a symmetric, non-swept wing. Typically, BLFs are placed on swept wings because there is significantly more spanwise flow when compared to non-swept wings. If BLFs truly provide benefits in aerodynamic efficiency because they restrict spanwise flow that contributes to wingtip vortex rollup, it is then obvious that non-swept wings would show less difference in aerodynamic efficiency with BLFs. This was reflected in the coefficient variations with respect to angle of attack in **Figure 8a**. Both the lift coefficient variations from the baseline and straight BLFs cases coincide with each other. Any changes in lift as a result of BLFs are within the uncertainty of the sensor, therefore it can be said that straight BLFs provide relatively no change in lift for the AR 4 NACA 0012 wing. This was a rather interesting finding considering all of the research that has shown BLFs to provide lift benefits. Those lift benefits, however, are only seen at high AoAs and typically on cambered, swept wings. Papers such as the one by Walker and Bons [10] test BLFs on cambered, swept wings until they stall and even past stall conditions. The AoA range considered in this paper only reached 15º, therefore post stall conditions are not investigated. Given the angle of attack range tested and the symmetrical, non-swept wing configuration, the similarity between the baseline and straight BLFs cases can be justified. **Figure 8b** shows the coefficient of drag data and the same thing is seen here. Similar to the lift coefficient, any moderate increment in drag is within the uncertainty of the sensor so it can be said that straight BLFs provide relatively no change in drag for the AR 4 NACA 0012 wing. Because both lift and drag do not change, aerodynamic efficiency of the wing with straight BLFs will be similar to the baseline NACA 0012 wing.

The slope of the curve for the coefficient of lift with respect to angle of attack is related to the AR of a wing. According to B.W. McCormick [11] this relationship can be calculated using McCormick's formula which is shown in **Equation 5**.

$$a = \frac{dC_L}{d\alpha} = a_0\left(\frac{AR}{AR + 2\dfrac{(AR+4)}{(AR+2)}}\right) \quad (\mathbf{5})$$

In this formula, $a_0 = 2\pi\ rad^{-1}$ which is the slope of a symmetric airfoil according to the Thin Airfoil Theory.

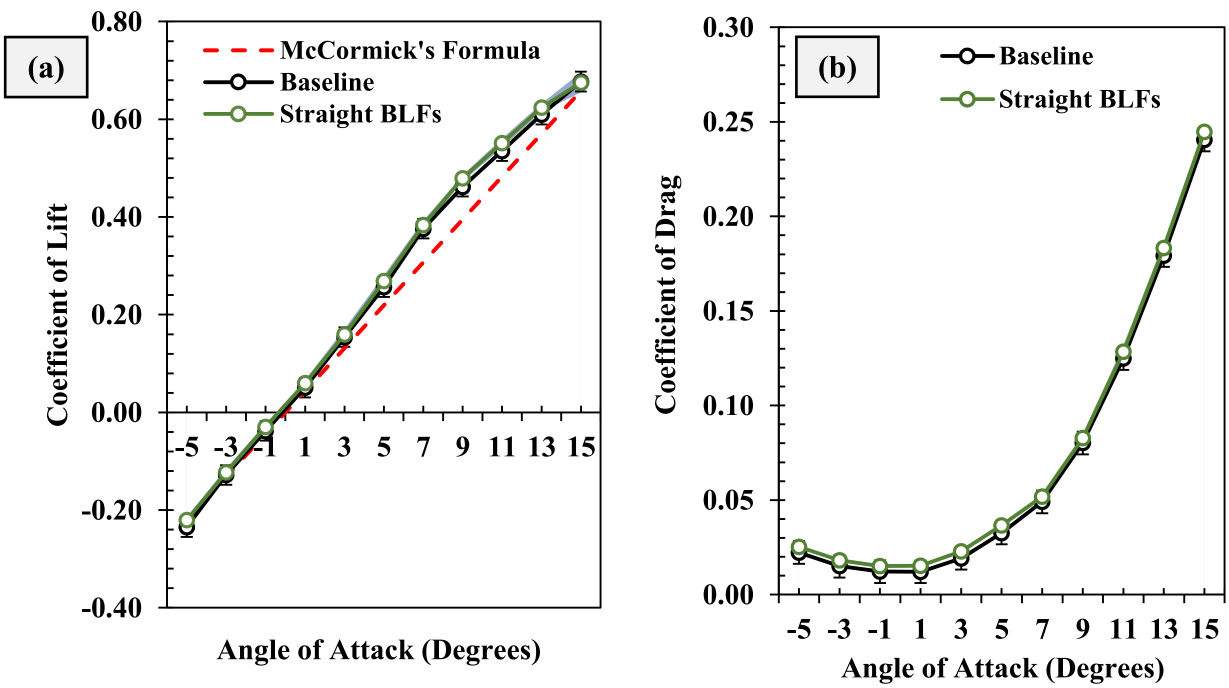

Figure 8: Straight BLF aerodynamic force data compared to the baseline wing; (a) coefficient of lift vs. AoA and (b) coefficient of drag vs. AoA.

When this equation was implemented using an AR of 4, however, the plot was off by a factor of two. Because of this, the baseline data better fits an AR of 2 which was the semispan of the test wings. This lift curve slope is represented by the red dotted line shown in the coefficient of lift vs. AoA plot for the straight BLF and baseline test configurations in **Figure 8a**. As stated earlier, the straight BLF test configurations differed in upper or lower surface coverage as well as spanwise location. The configuration seen below in **Figure 9a** is the "upper middle" configuration which means that the BLF is located on the upper surface of the wing and in the middle of the span. **Figure 9b** shows the baseline wing test configuration for comparison.

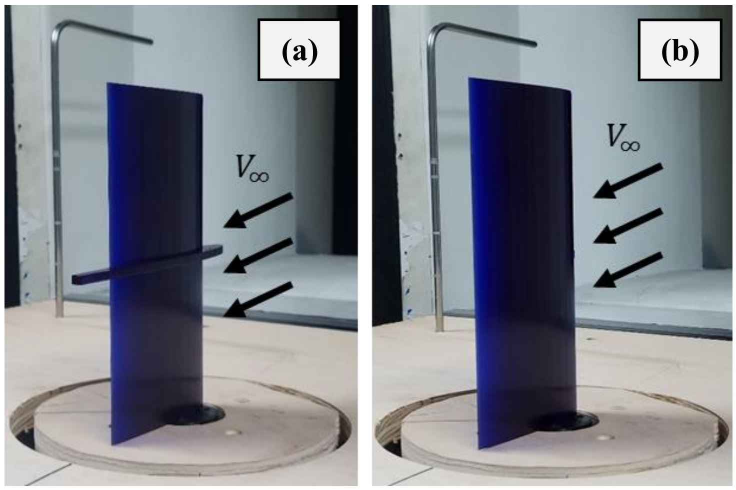

Figure 9: (a) One of four straight BLF test configurations (upper surface, middle of the wing) along with (b) the baseline wing test configuration.

3.3 Sensitivity of CBLFs to Fractional Spanwise Coverage

This section will discuss the sensitivity of CBLFs to fractional spanwise coverage. Here, half span CBLFs will be compared to quarter span CBLFs. To observe the differences between the two, **Figure 10a** shows a half span test configuration while **Figure 10b** shows a quarter span test configuration in the same orientation; located on the upper surface near the wingtip and curving inward.

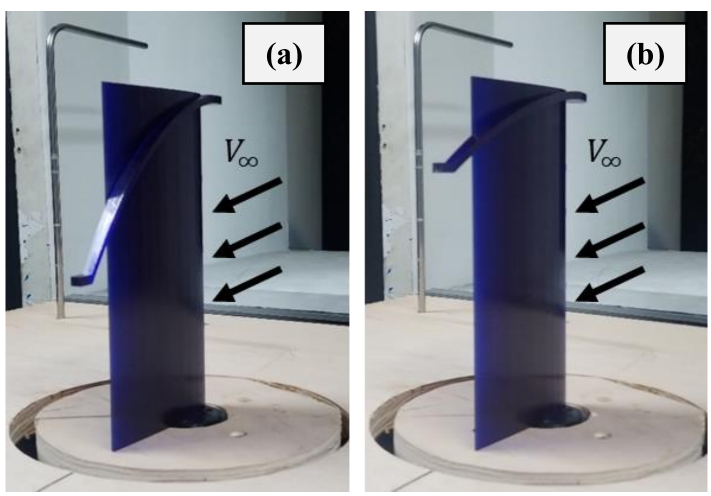

Figure 10: (a) Half span CBLF vs. (b) quarter span CBLF with the same orientation (upper wingtip inward).

The lift and drag coefficient plots in this section feature lines as well as shaded regions. The lines indicate the average values of the data series that correspond to a certain general configuration. For example, in the half span plots, the solid line represents the average of all 8 half span CBLF test configurations (upper and lower surface: middle inward, middle outward, wingtip inward, wingtip outward). The shaded region on the same plot does not represent standard deviations, but instead the spread of data among all eight configurations of the half span CBLF.

Any changes in coefficient of lift for all CBLF test configurations when compared to the baseline wing are within the uncertainty of the sensor. This is similar to the straight BLF test configurations and the reasoning for this may be the same. Like the straight BLFs, CBLFs were not tested past stall conditions and the same non-swept, symmetric wing was used. The plots of coefficient of lift vs. AoA for the half span and quarter span CBLFs are shown in **Figure 11a** and **Figure 11b** respectively.

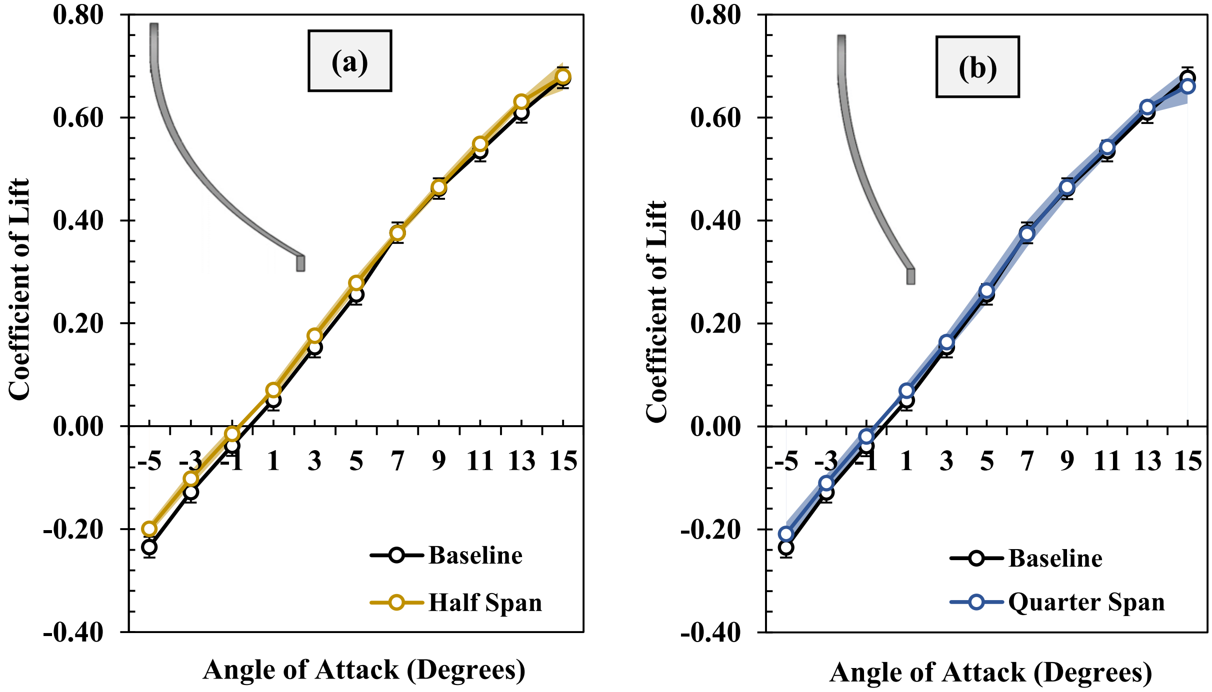

Figure 11: Coefficient of lift data compared to baseline for CBLFs with different fractional spanwise coverages; (a) half span CBLFs and (b) quarter span CBLFs. Shaded regions represent the spread of data among the eight test configurations that were plotted together.

Although the effects of CBLFs are not apparent in the coefficient of lift data, they are present in the coefficient of drag data. Each CBLF, although having a higher drag coefficient at low AoAs, experiences lower drag at AoAs beginning between 10º and 11º when compared to the baseline. The fact that CBLFs begin to show signs of improved performance only at high AoAs is consistent with previous research that has shown the same [9], [10]. The difference here, however, is that the drag coefficient changed while the lift remained relatively the same indicating that the parasitic drag of the wing was reduced. It is important to note that *all* CBLF test configurations show lower drag than the baseline wing at AoAs beginning between 10º and 11º. This result shows conclusive evidence that irrespective of the curvature or other orientation, CBLFs will perform better than a wing without CBLFs at high AoAs. Further, since the baseline wing and the wing with straight BLFs exhibit relatively the same performance, these results also show that CBLFs perform better than straight BLFs at high AoAs.

When comparing the performance of the two CBLFs individually, the data shows that the quarter span CBLFs generally experience less drag than the half span CBLFs. This shouldn't be surprising considering the half span CBLFs introduce more wetted area into the flow. At 13º AoA, the average of all eight quarter span CBLF test configurations shows a 25% increase in aerodynamic efficiency when compared to the baseline. At the same AoA, the average of all eight half span CBLF test cases show an 18% increase in aerodynamic efficiency. Based on this data, and the coefficient of drag data, it is safe to say that quarter span CBLFs generally perform better than half span CBLFs. **Figure 12a** shows the coefficient of drag vs. AoA data for the both the half span and quarter span CBLFs. **Figure 12b** shows the aerodynamic efficiency vs. positive AoAs for these two different CBLFs. Again, the solid curves represent the average of all eight test configurations while the shaded regions show the spread between the test cases.

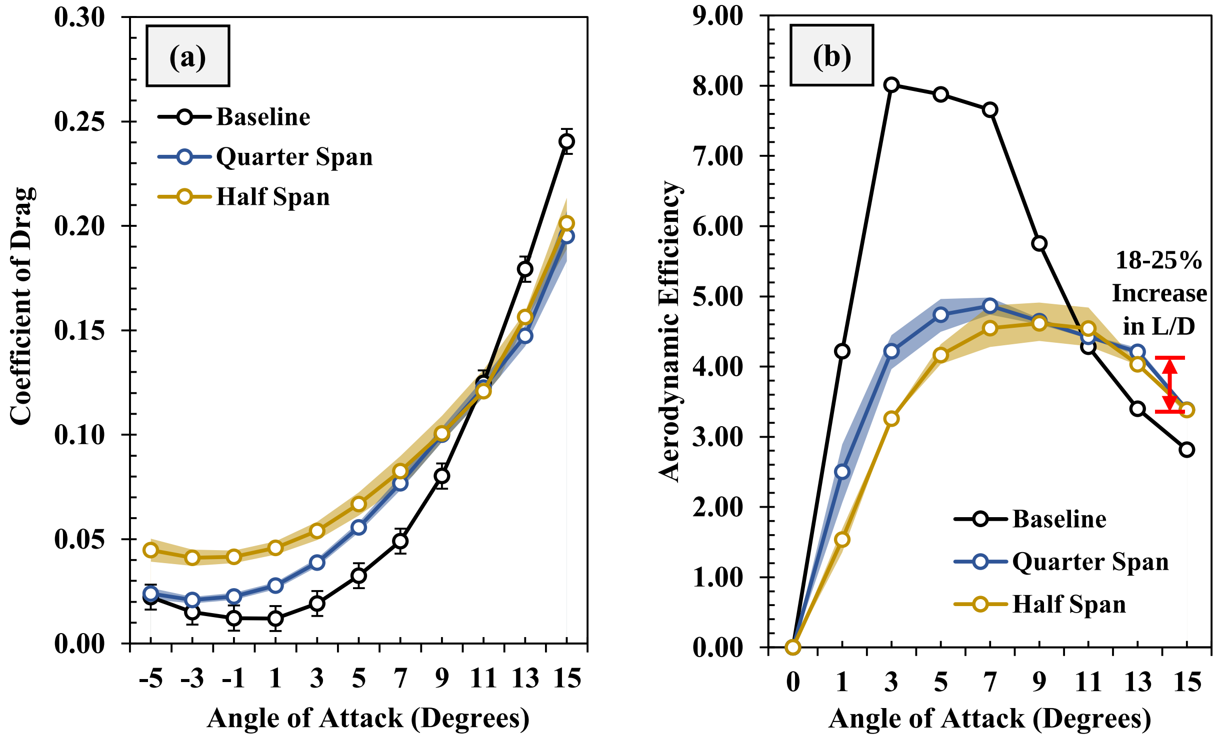

Figure 12: (a) Coefficient of drag data compared to baseline for CBLFs with different fractional spanwise coverages; half span CBLFs and quarter span CBLFs. (b) Aerodynamic efficiency vs. positive AoAs. Again, shaded regions represent the spread of data among the eight test configurations that were plotted together.

3.4 Sensitivity of CBLFs to Upper/Lower Surface Coverage

As stated earlier, any changes in the lift coefficient data for every CBLF test configuration lie within the uncertainty of the force sensor. For this reason, lift coefficient data will not be mentioned in the remainder of the paper. Instead, effects of CBLFs on aerodynamic efficiency will be explained by changes in the drag coefficient data alone. Traditionally, BLFs are placed on the upper surface of a wing. The idea of testing CBLFs on both surfaces independently is not common practice and has not been done before to the best of the authors' knowledge. Nevertheless, variation between the upper and lower surface CBLF was seen for the half span case. For the quarter span CBLF, however, almost no variation was observed in the data when comparing quarter span CBLFs on the upper and lower surfaces which is interesting.

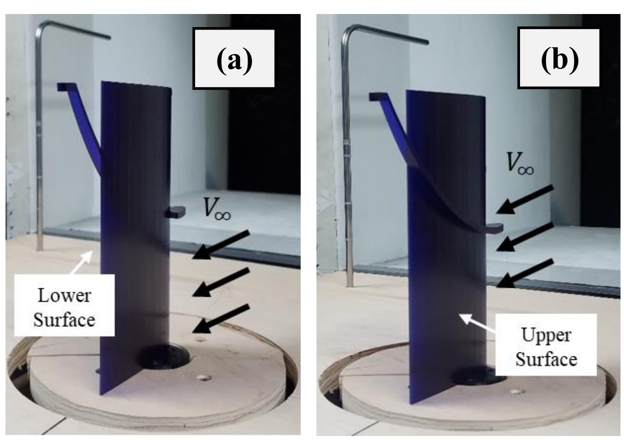

Figure 13: Half span CBLF on (a) lower surface and on (b) upper surface with the same orientation.

As a result, variation in aerodynamic efficiency is not significantly affected by placing quarter span CBLFs the upper or lower surface. Regardless, the quarter span CBLF provided greater drag reduction benefits irrespective of whether or not they are in the upper or lower surface when compared to all of the half span configurations. For the half span CBLFs though, variation was seen. **Figure 13a** and **Figure 13b** show the half span CBLF in the same test configuration but on the opposite surface in the same orientation (wingtip outward).

As freestream air passes over a wing, flow on the upper surface curves inwards towards the wing root while flow on the lower surface curves outwards towards the wingtip. This is due to pressure difference that occurs at the wingtip causing the higher pressure air on the lower surface to roll up onto the upper surface creating wingtip vortices. In theory, variations in aerodynamic coefficients should be seen by placing CBLFs on the upper or lower surface because spanwise flow is moving in opposite directions on these two surfaces. On the upper surface, spanwise flow migrates towards the root and on the lower surface, it migrates towards the wingtip.

Drag force data in **Figure 14a** shows that more drag is generated when CBLFs are positioned on the lower surface of the wing within the AoA range of 1º to 13º. After 13º however, CBLFs operate similarly on the upper and lower surfaces. Aerodynamic efficiency data in **Figure 14b** shows that at 11º AoA, CBLFs are about 9% more aerodynamically efficient than the baseline when placed on the lower surface of the wing. At the same AoA, CBLFs improve aerodynamic efficiency by about 20% when placed on the upper surface. This conclusively shows that adding CBLFs to the upper surface provides greater reduction in drag coefficient when compared to adding the CBLF to the lower surface.

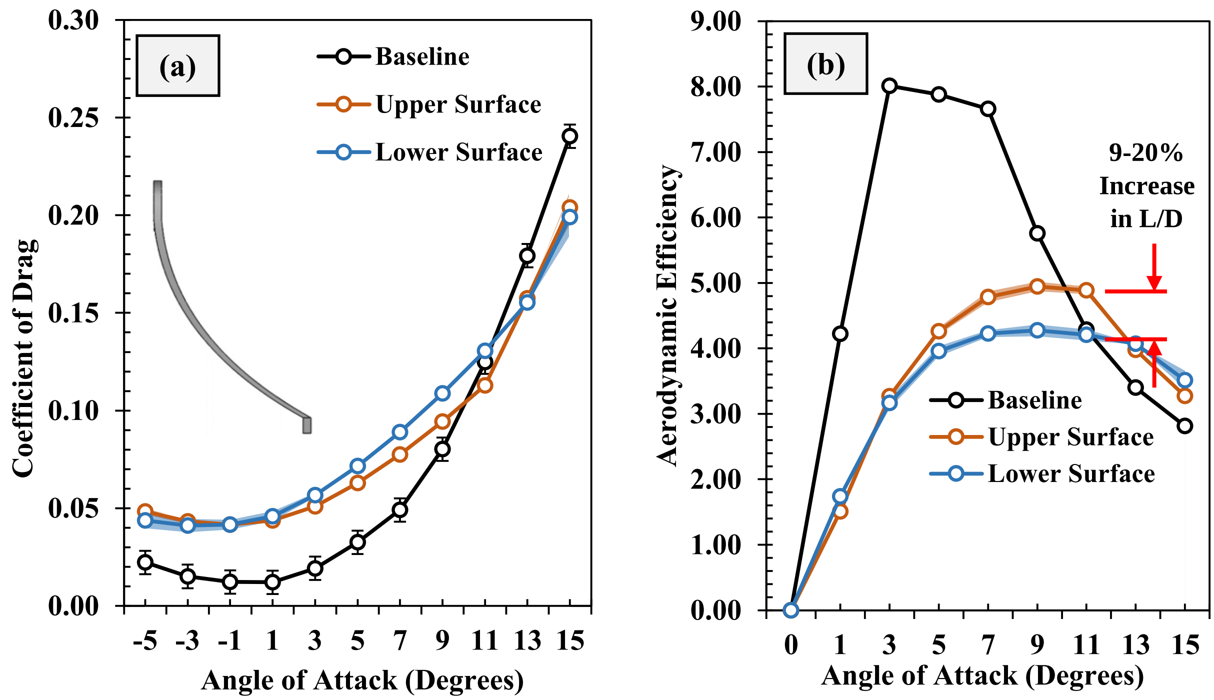

Figure 14: (a) Coefficient of drag data for half span CBLFs located on the upper or lower surface shows that half span CBLFs generate more drag on the lower surface between 1º and 13º AoA. (b) Aerodynamic efficiency vs. positive AoA data shows half span CBLFs are generally more aerodynamically efficient on the upper surface.

3.5 Flow Visualization

Mini tuft flow visualization was performed on the NACA 0012 wing with CBLFs covering both the upper and lower surfaces of the wing. The CBLFs covering both surfaces showed similar lift and drag characteristics as shown in **Figure 14**. These CBLFs were tested in similar configurations which differed in fractional spanwise coverage (half, one-third, and quarter span), spanwise location (middle and wingtip), and direction of curvature (inward and outward). When the baseline wing was placed at a higher angle of attack, the flow separation can be seen clearly in **Figure 15a**. However, the results from the flow visualization showed that flow remains attached to some areas of the upper surface due to the presence of the CBLF. The flow separation in the baseline case seen by turbulent motion of the tufts indicated that the stall angle was reached. **Figure 15b-d** shows the three CBLF test cases that recorded the highest increase in aerodynamic efficiency out of all the test cases. These images were taken at the moment that flow begins to detach from the upper surface and red circles and arrows point to areas where this occurs. It can clearly be seen that the CBLFs keep flow attached at more locations however the CBLF test cases are at a higher AoA than the baseline wing when flow begins to become detached. The one-third span wingtip inward test configuration (**Figure 15d**) showed the most flow to remain attached once the wing begins to stall.

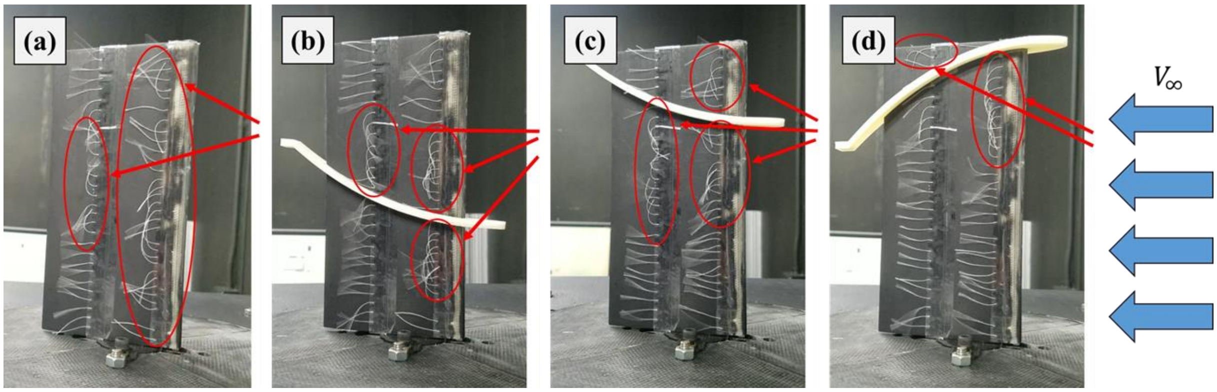

Figure 15: Mini tuft flow visualization on a previous CBLF test design that covered both wing surfaces simultaneously showed CBLFs to keep flow attached at higher AoAs than the baseline and in more locations. (a) Baseline wing shows that as flow separates everywhere as flow detaches. (b) Quarter span middle outward, (c) quarter span wingtip outward, (d) one-third span wingtip inward.

The results from force based experiments clearly show variation in aerodynamic efficiency with the use of CBLFs. Any positive variation as a result of CBLFs, however, is only seen at high AoAs so having CBLFs at lower AoAs would be detrimental to wing performance. Therefore, in order to utilize the benefits of both straight BLFs and CBLFs, a morphing boundary layer fence could be a viable solution to increase the overall aerodynamic efficiency of the wing. This idea is currently being explored by the authors. A morphing BLF will be designed to adjust its curvature during flight to feature the optimum curvature at any point. The BLF will be actuated at high AoAs to have higher curvature, minimizing drag during takeoff and landing phases of flight. During cruise conditions, the BLF would be actuated to be straight at low AoAs. A schematic of this type of actuation is shown in **Figure 16**.

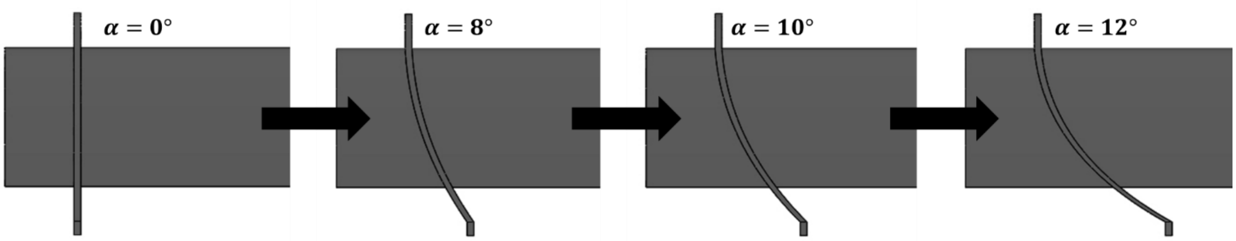

Figure 16: Concept of a morphing BLF that can adjust its curvature during flight to utilize straight BLFs at low AoAs (cruise) and CBLFs at high AoAs (takeoff and landing).

Another potential application of these CBLFs is that they can act as adaptive control surfaces. Roll, pitch and yaw maneuvers can be achieved through symmetric and asymmetric actuation of these BLFs in the wing and in the horizontal tail of the aircraft which will be explored in the follow up paper.

IV. Future Work

Although significant variation is seen in the aerodynamic efficiency of a NACA 0012 wing with the use of CBLFs, more research is necessary to quantify the surface flow changes and wingtip vortex changes. Force based testing revealed key sensitivities with respect to CBLF placement and size which will be further explored through surface flow visualization as well as Particle Image Velocimetry (PIV). PIV will also be performed on the wingtip vortex to understand how the wingtip vortex is effected by CBLFs. CBLFs will also be tested on wings with cambered airfoils and sweep.

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